# Fourier Transform

# Jean Baptiste Joseph Fourier

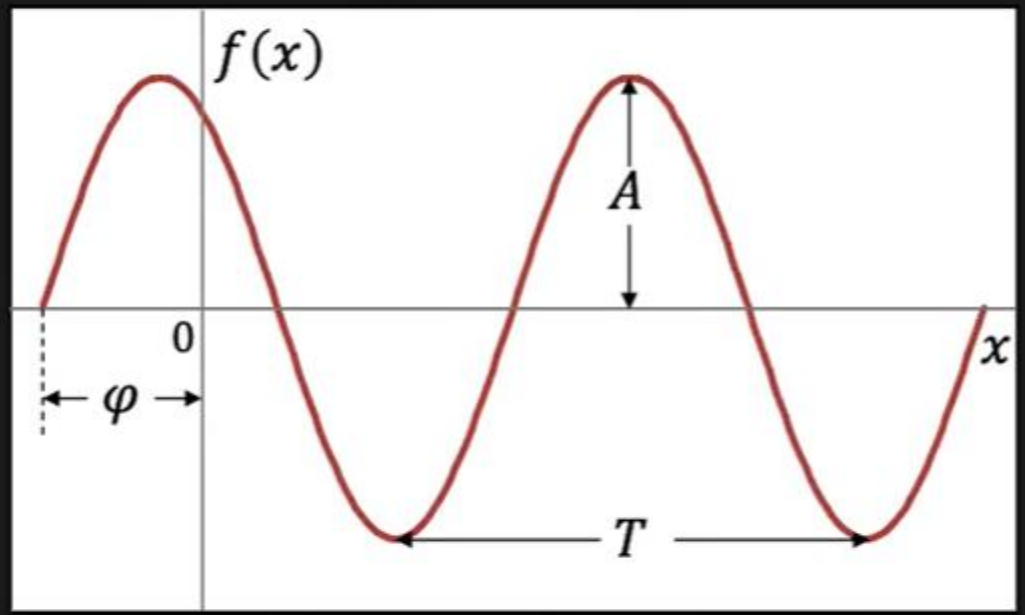



(1768-1830)

Any Periodic Function can be rewritten as a Weighted Sum of Infinite Sinusoids of Different Frequencies.

# Sinusoid

$$f(x) = A\sin(2\pi ux + \varphi)$$



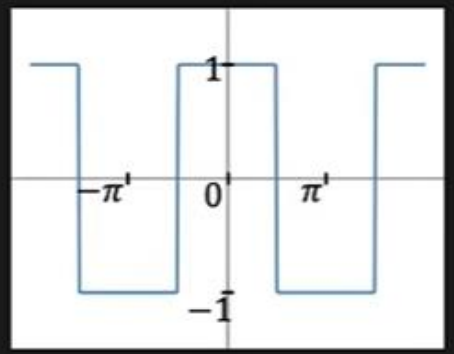

$A$: Amplitude

$T$: Period

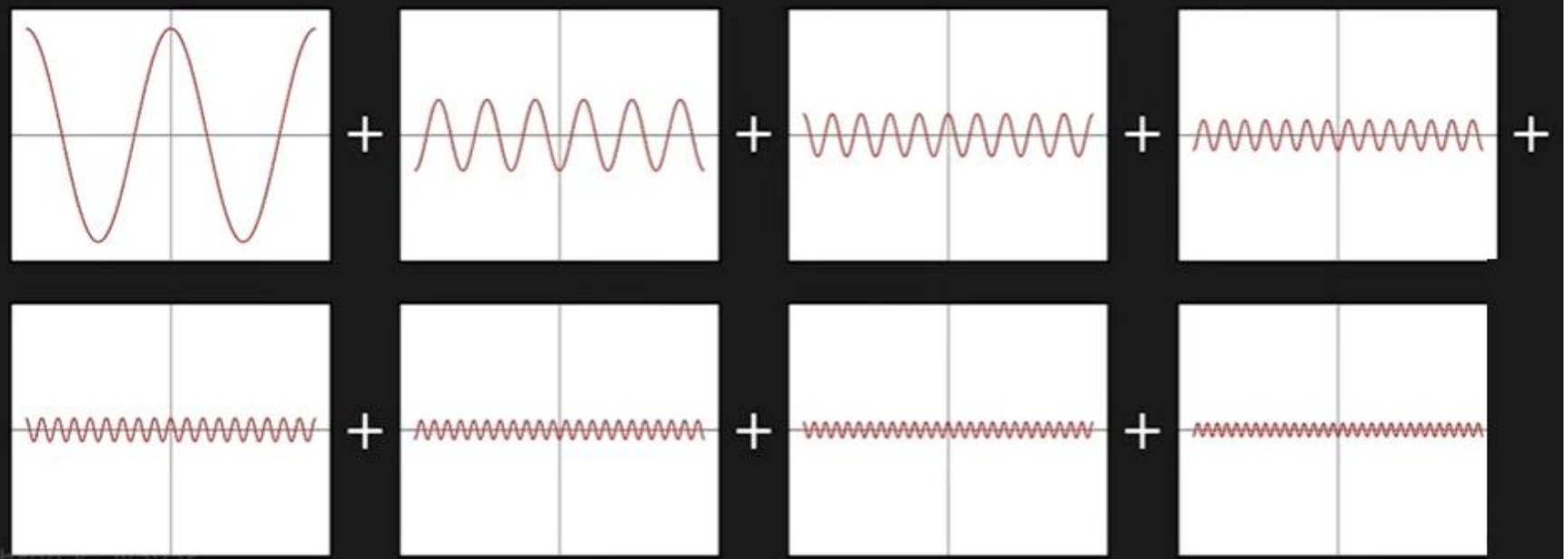
$\varphi$: Phase

$u$: Frequency $(1/T)$

# Fourier Series

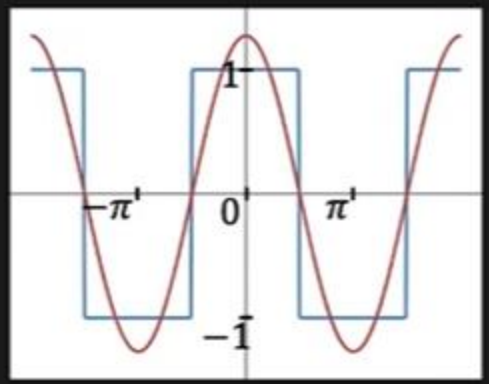



Square Wave
(Period $2\pi$)



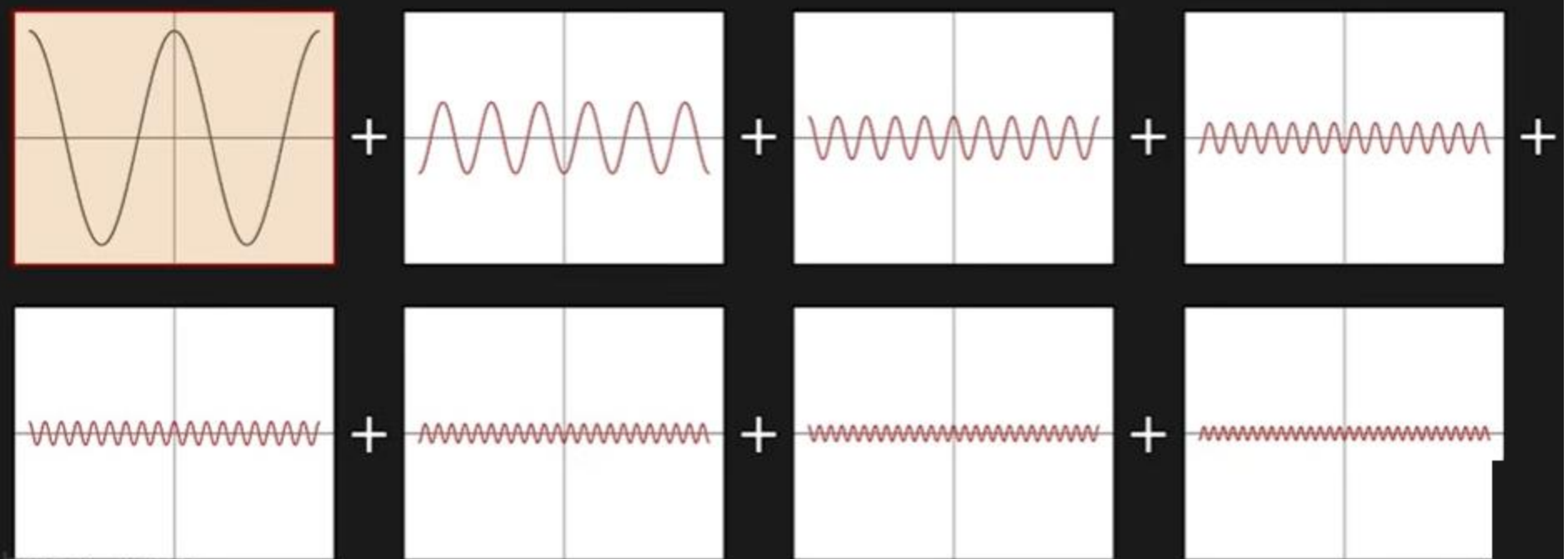

20 Shree K. Nayar

# Fourier Series

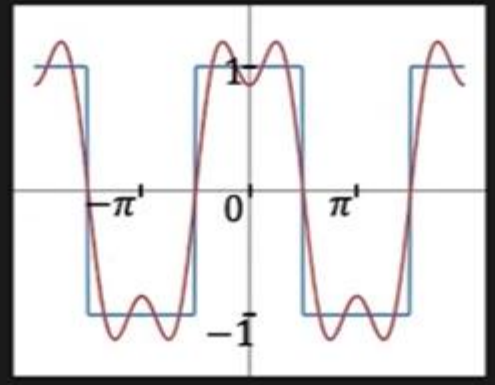



Square Wave
(Period $2\pi$)

Sum of Sinusoids

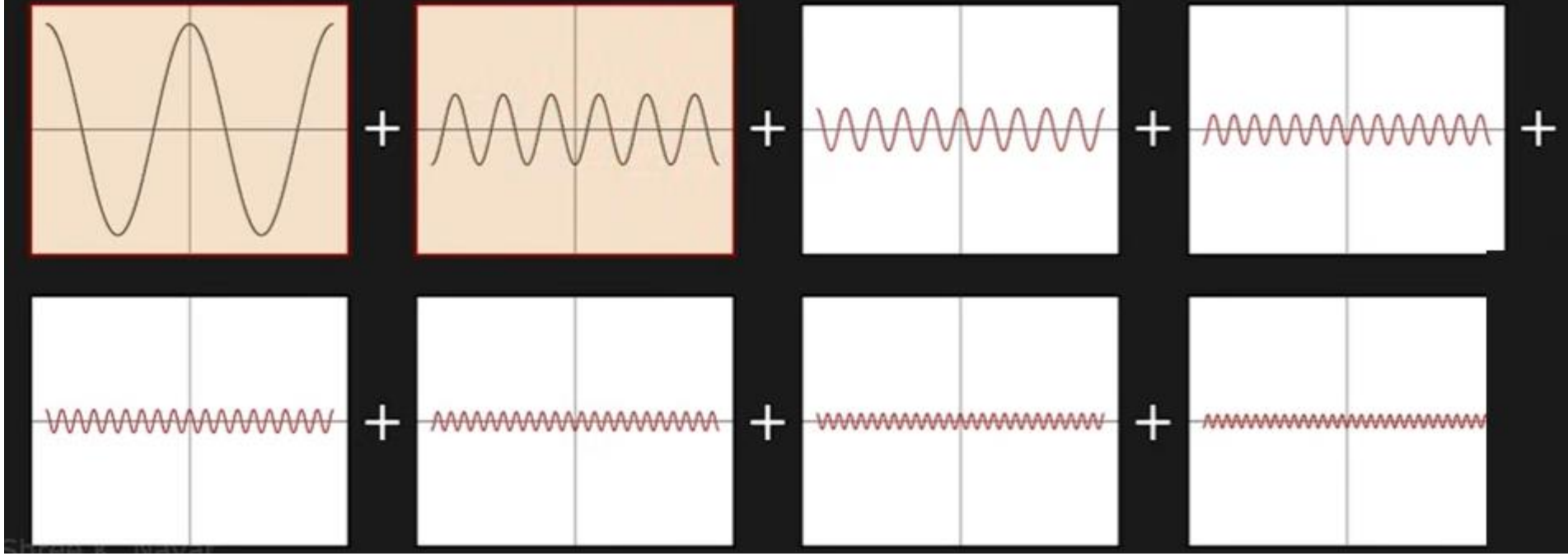


# Fourier Series

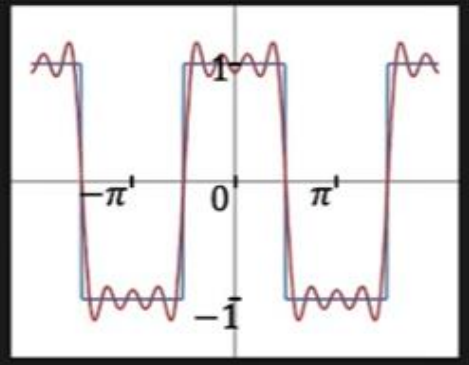



Square Wave
(Period $2\pi$)

Sum of Sinusoids

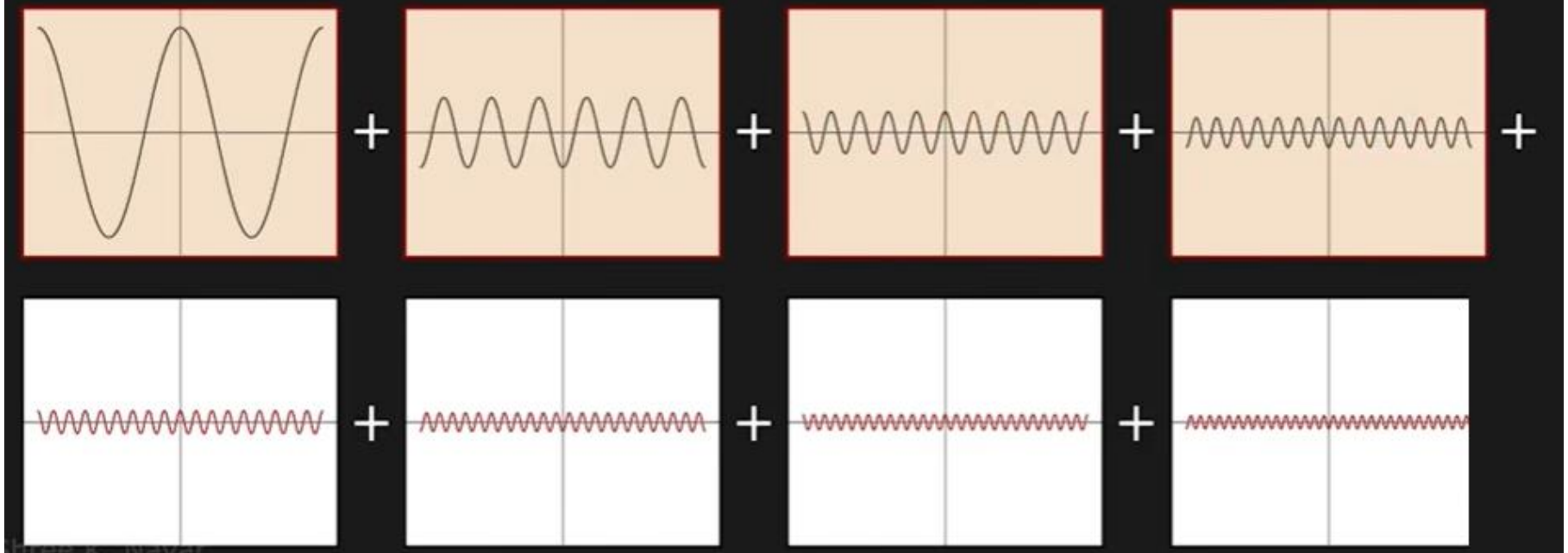


# Fourier Series

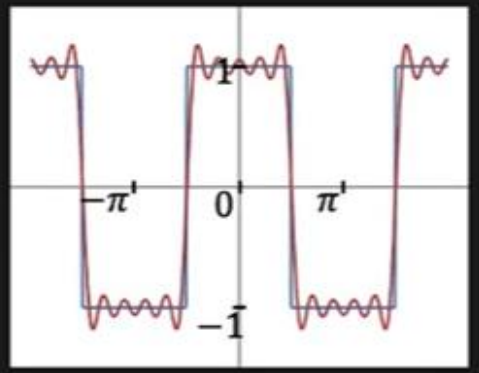



Square Wave
(Period $2\pi$)

Sum of Sinusoids

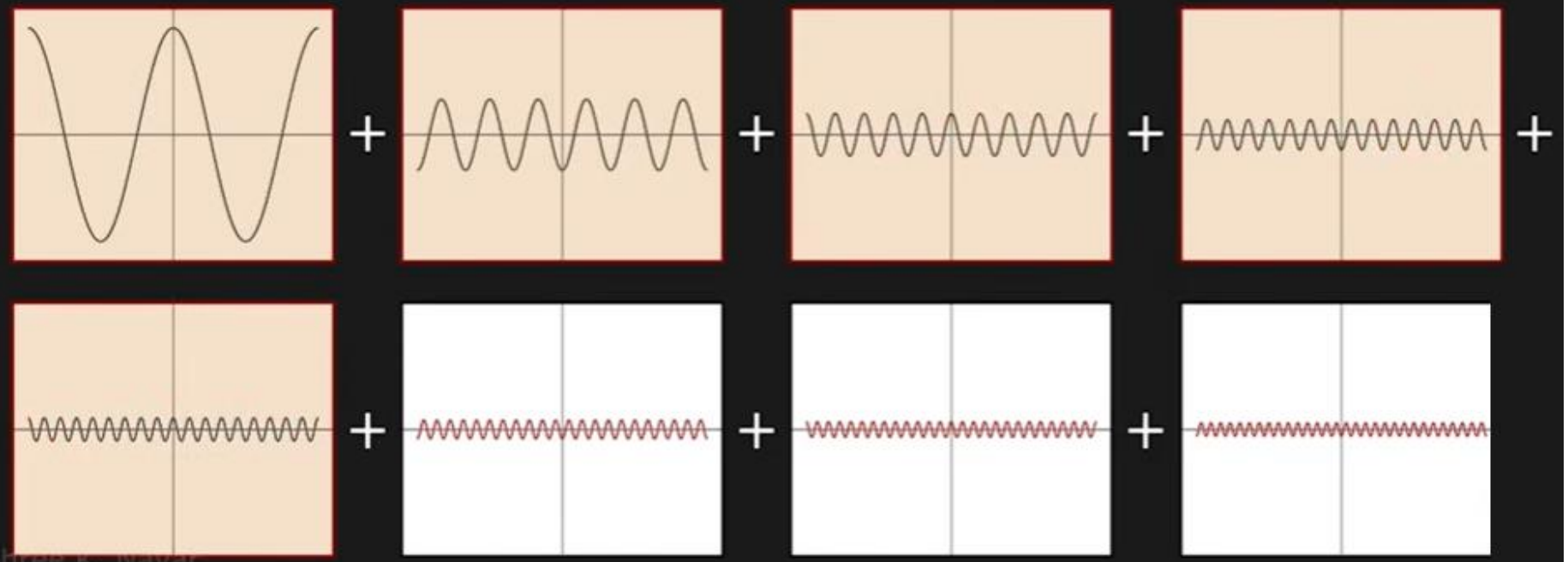


# Fourier Series



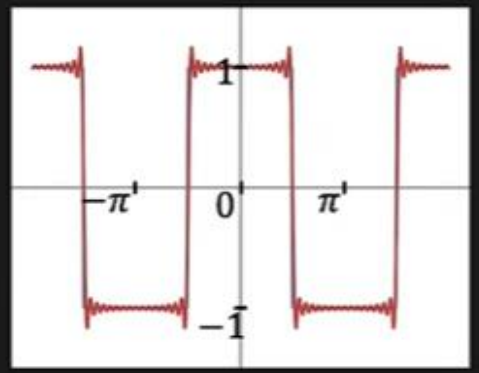

Square Wave
(Period $2\pi$)

Sum of Sinusoids



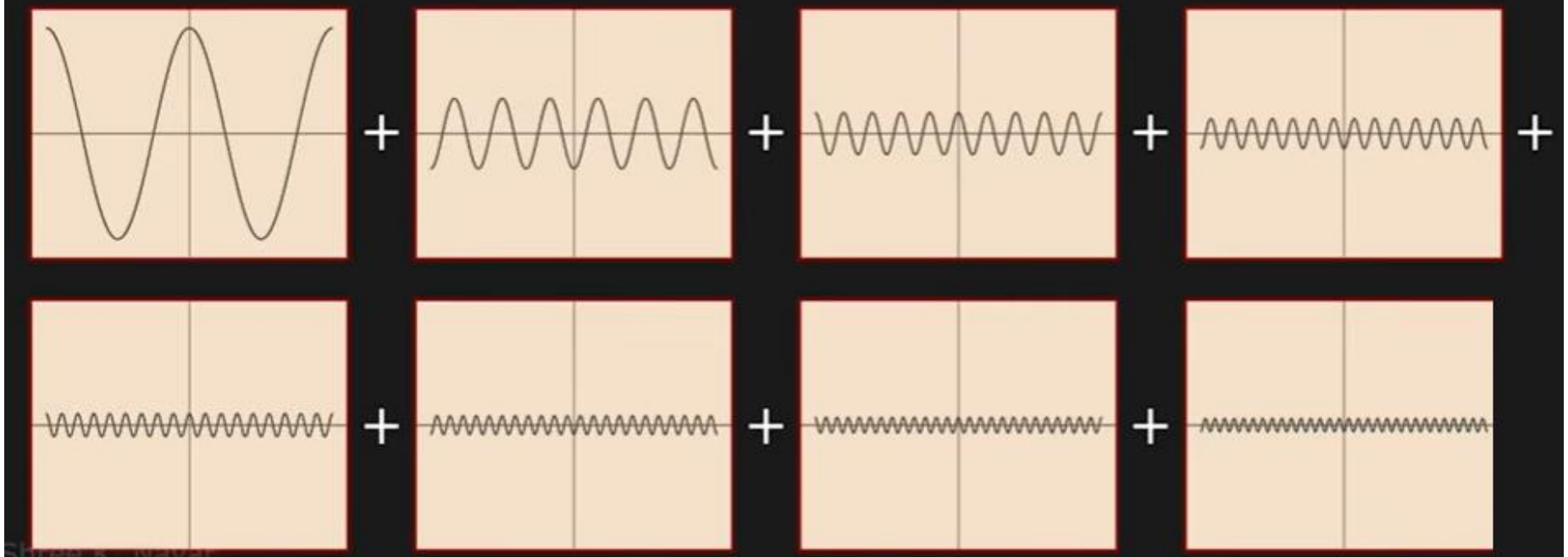
020 Shree K. Nayar

# Fourier Series



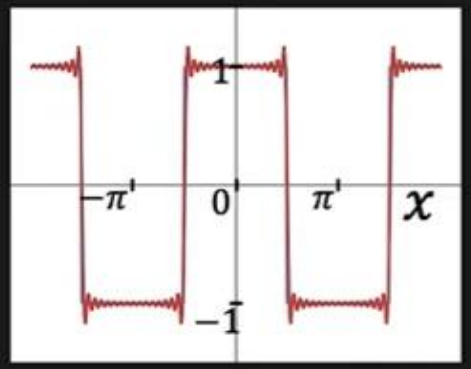

Square Wave
(Period $2\pi$)

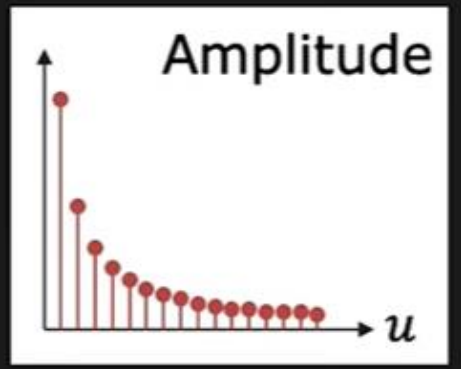
Sum of Sinusoids

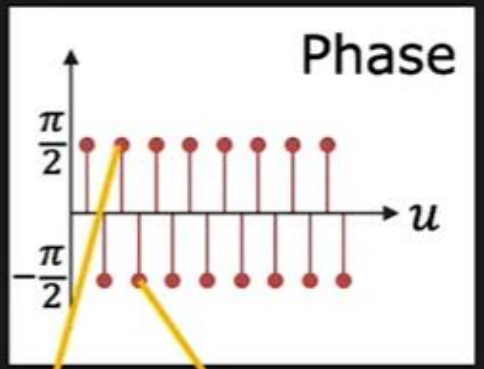


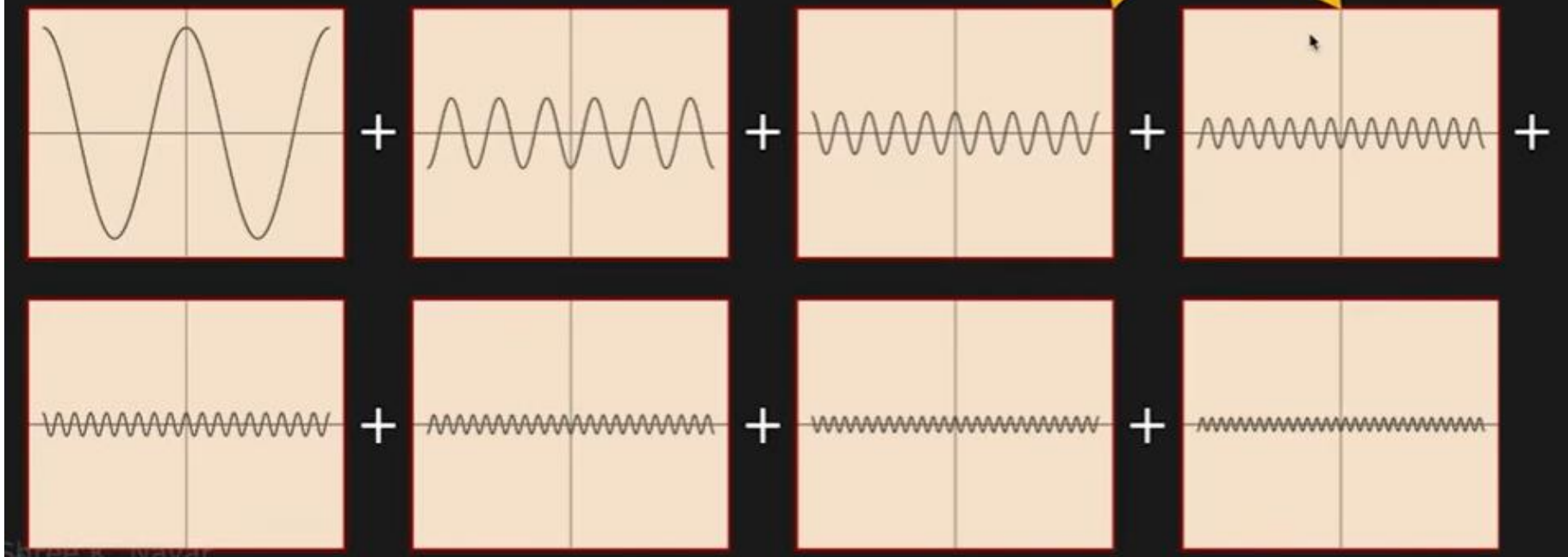
Shree K. Nayar

# Frequency Representation of Signal

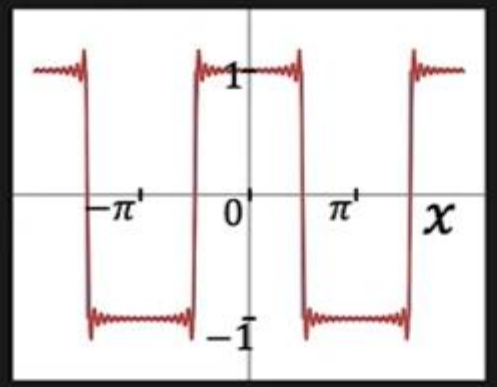


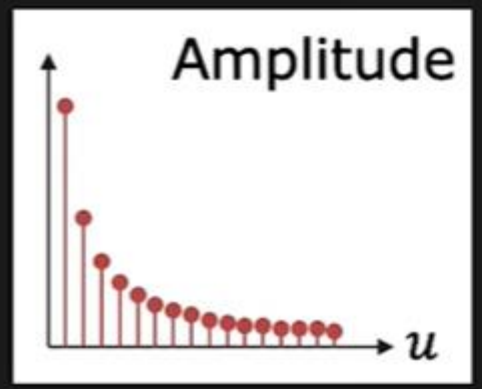



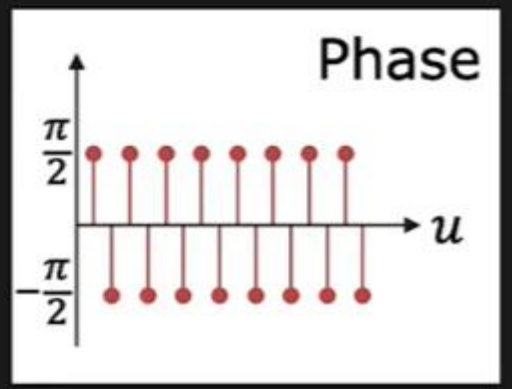



# Fourier Transform (FT)



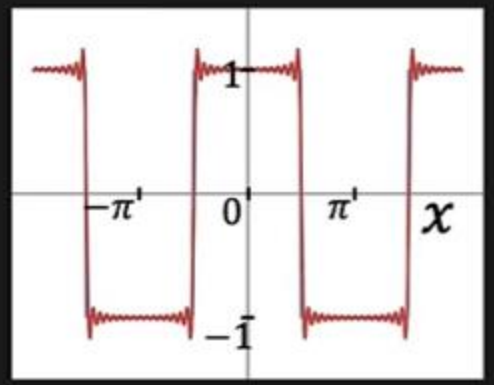



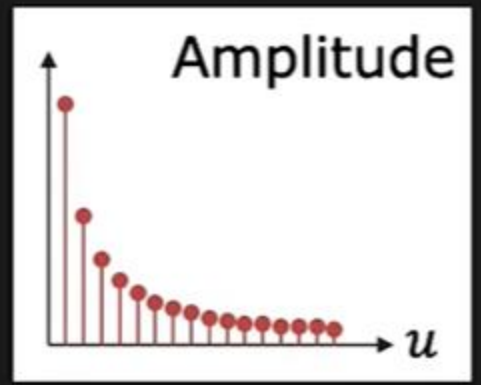



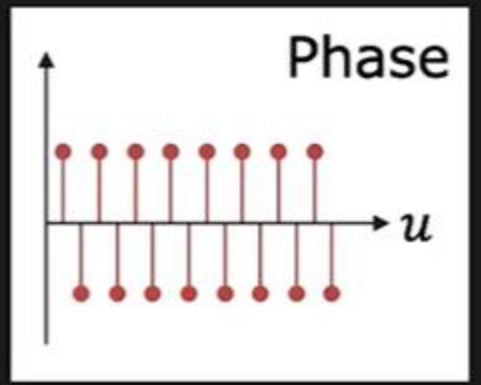

Represents a signal $f(x)$ in terms of Amplitudes and Phases of its Constituent Sinusoids.

$$f(x) \rightarrow \boxed{\text{FT}} \rightarrow F(u)$$

# Inverse Fourier Transform (IFT)




Computes the signal $f(x)$ from the Amplitudes and Phases of its Constituent Sinusoids.

$f(x)$ ← IFT ← $F(u)$

# Finding FT and IFT

Fourier Transform:

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi ux} dx$$

$x$: space

$u$: frequency

$e^{i\theta} = \cos\theta + i\sin\theta$

$i = \sqrt{-1}$

Inverse Fourier Transform:

$$f(x) = \int_{-\infty}^{\infty} F(u) e^{i2\pi ux} du$$

# Complex Exponential (Euler Formula)

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta} \qquad i = \sqrt{-1}$$

Expand $e^{i\theta}$ using Taylor Series:

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \frac{(i\theta)^6}{6!} + \dots$$

$$e^{i\theta} = \underbrace{\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \cdots\right)}_{\cos\theta} + i\underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots\right)}_{\sin\theta}$$

# Fourier Transform is Complex!

$F(u)$ holds the Amplitude and Phase of the sinusoid of frequency $u$.

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi ux} dx$$

$$F(u) = \mathfrak{Re}\{F(u)\} + i\,\mathfrak{Im}\{F(u)\}$$

# Fourier Transform is Complex!

$F(u)$ holds the Amplitude and Phase of the sinusoid of frequency $u$.

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi ux} dx$$

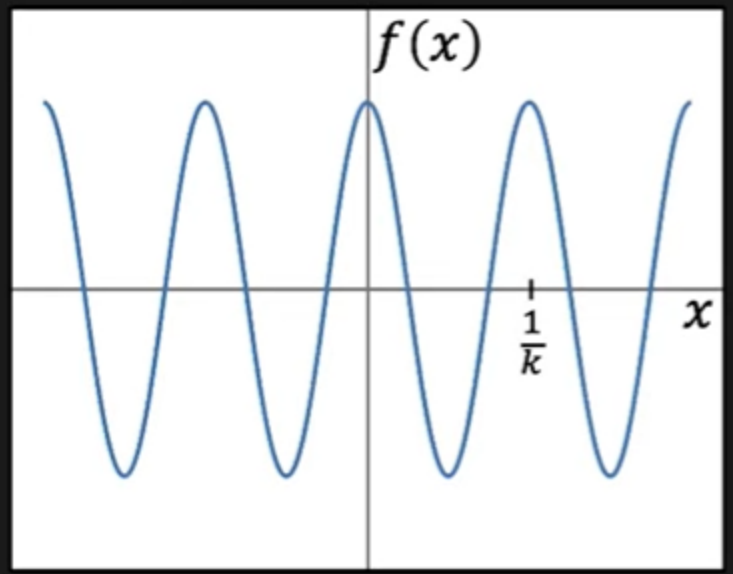
$$F(u) = \Re\mathrm{e}\{F(u)\} + i\,\Im\mathrm{m}\{F(u)\}$$

Amplitude: $A(u) = \sqrt{\Re\mathrm{e}\{F(u)\}^2 + \Im\mathrm{m}\{F(u)\}^2}$

Phase: $\varphi(u) = \mathrm{atan2}(\Im\mathrm{m}\{F(u)\}, \Re\mathrm{e}\{F(u)\})$

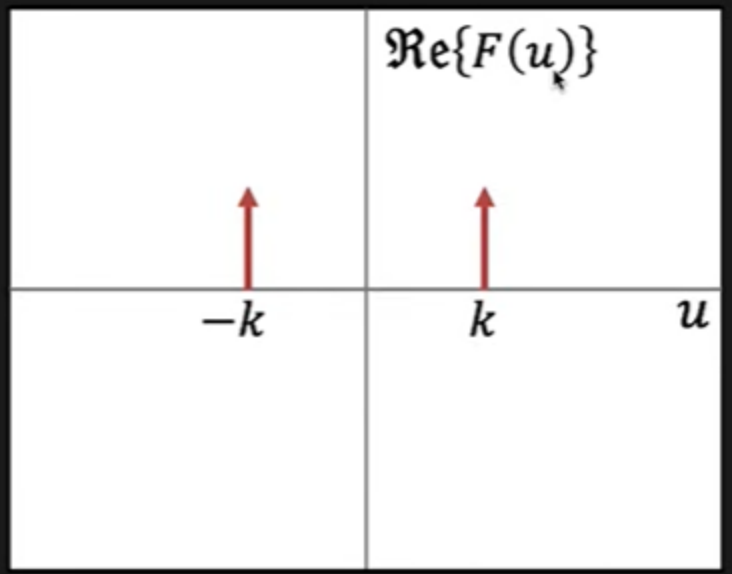
# Fourier Transform Examples

## Signal $f(x)$




$$f(x) = \cos 2\pi kx$$

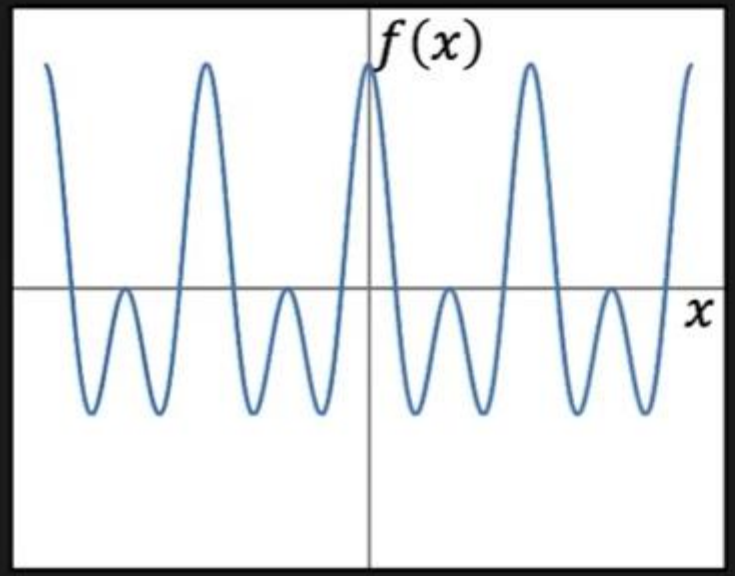
## Fourier Transform $F(u)$




$$F(u) = \tfrac{1}{2}[\delta(u+k) + \delta(u-k)]$$

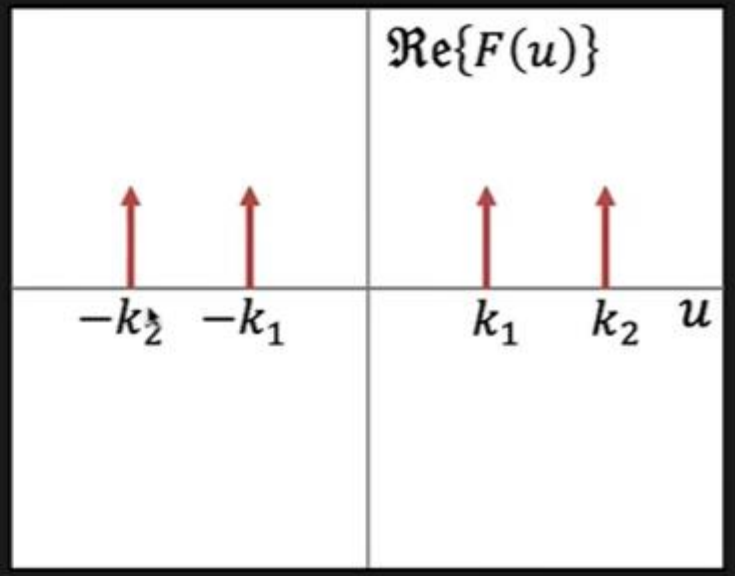
# Fourier Transform Examples

## Signal $f(x)$




$$f(x) = \cos 2\pi k_1 x + \cos 2\pi k_2 x$$

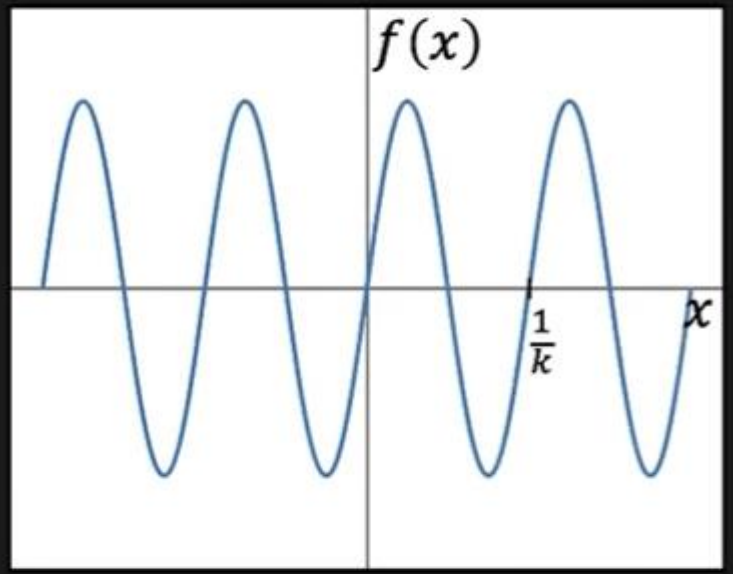
## Fourier Transform $F(u)$




$$F(u) = \tfrac{1}{2}\left[\delta(u + k_1) + \delta(u - k_1) + \delta(u + k_2) + \delta(u - k_2)\right]$$

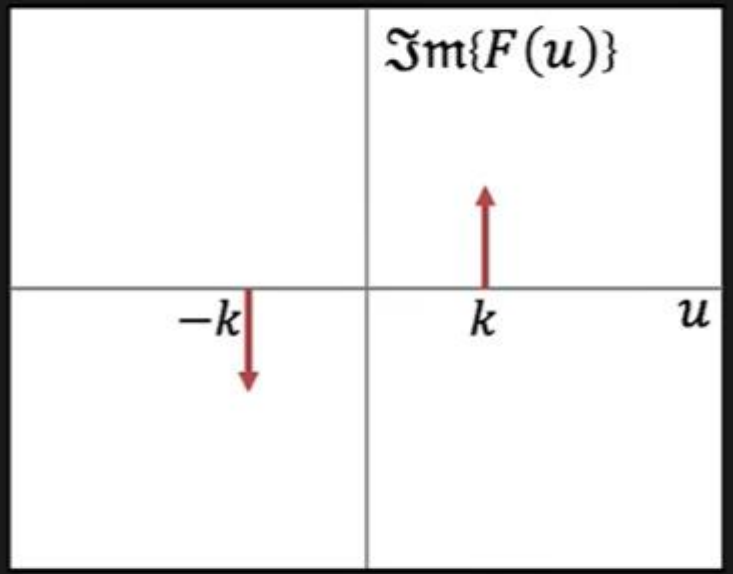
# Fourier Transform Examples

## Signal $f(x)$




$$f(x) = \sin 2\pi kx$$

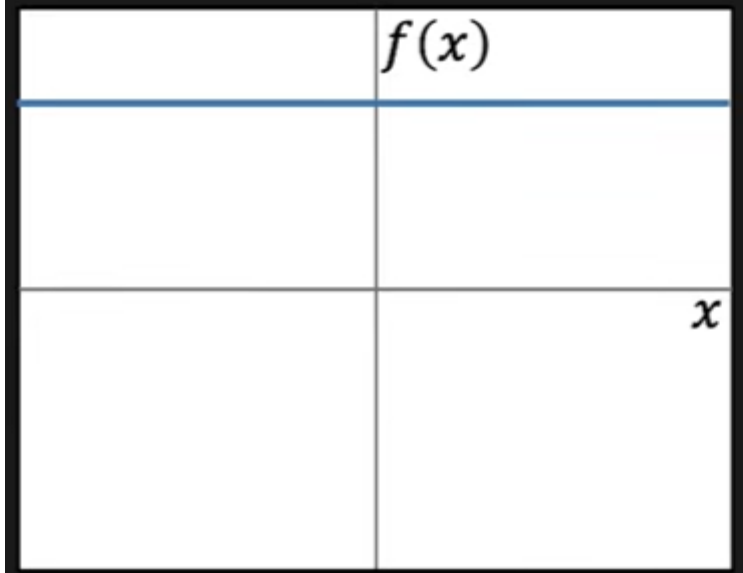
## Fourier Transform $F(u)$




$$F(u) = \tfrac{1}{2}i[\delta(u+k) - \delta(u-k)]$$

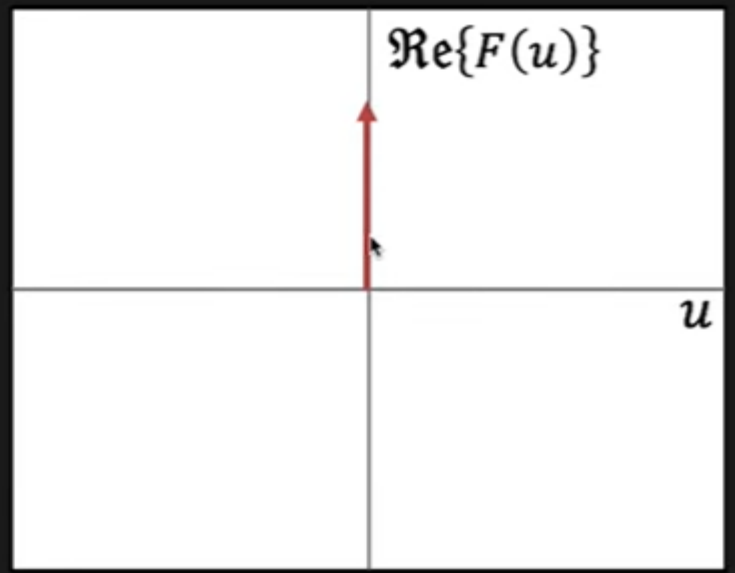
# Fourier Transform Examples

Signal $f(x)$




$f(x) = 1$

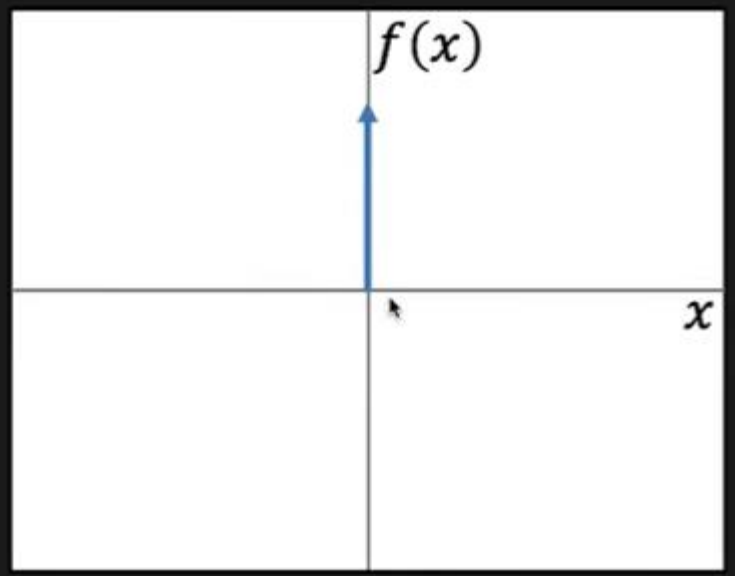
Fourier Transform $F(u)$




$F(u) = \delta(u)$

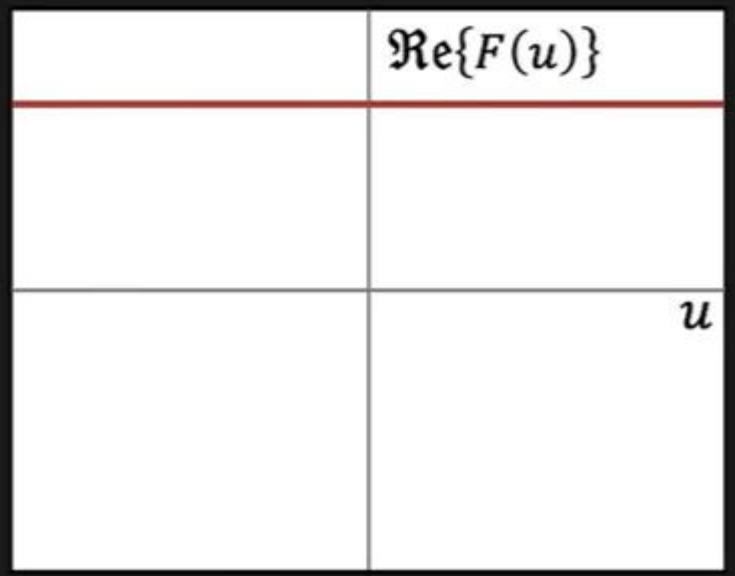
# Fourier Transform Examples

## Signal $f(x)$




$$f(x) = \delta(x)$$

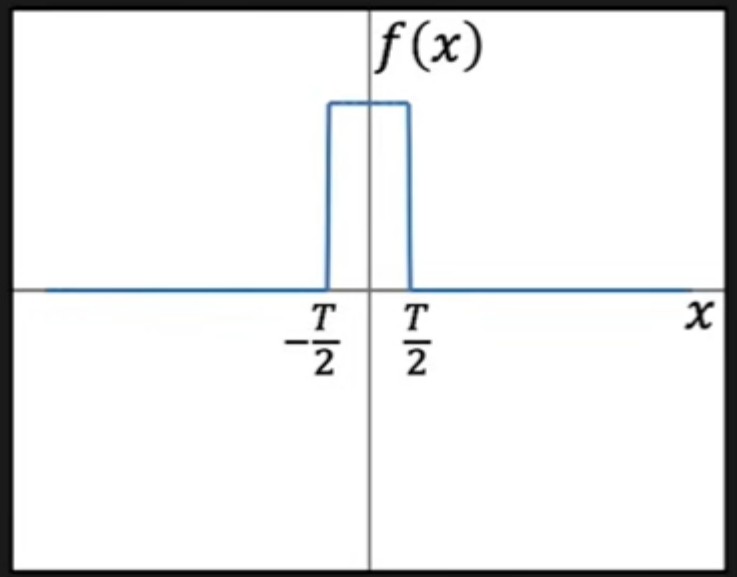
## Fourier Transform $F(u)$




$$F(u) = 1$$

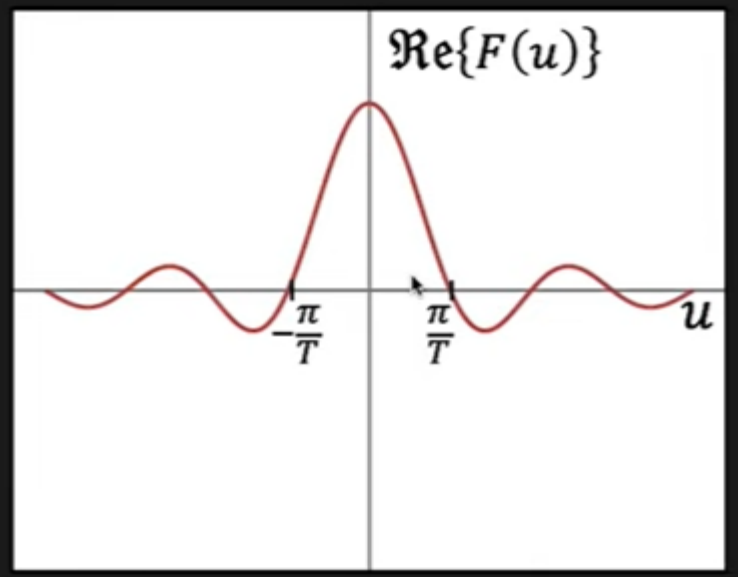
# Fourier Transform Examples

## Signal $f(x)$




$$f(x) = \text{Rect}\left(\frac{x}{T}\right)$$

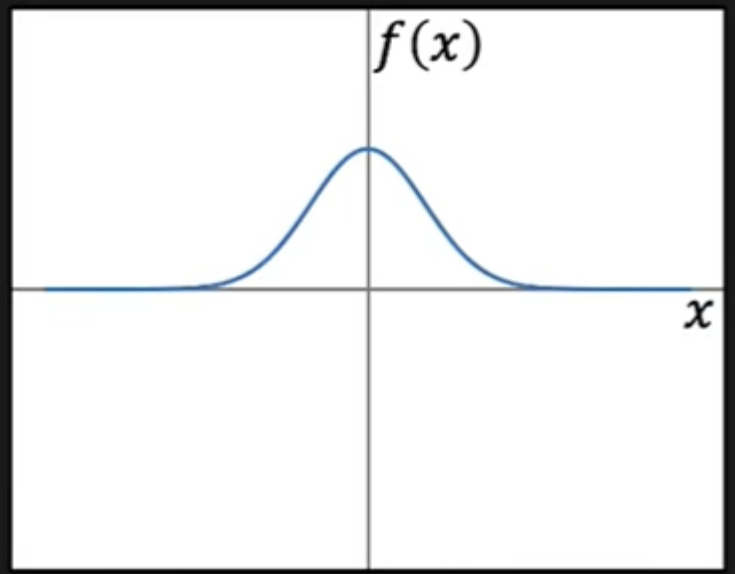
## Fourier Transform $F(u)$




$$F(u) = T\,\text{sinc}\,Tu$$

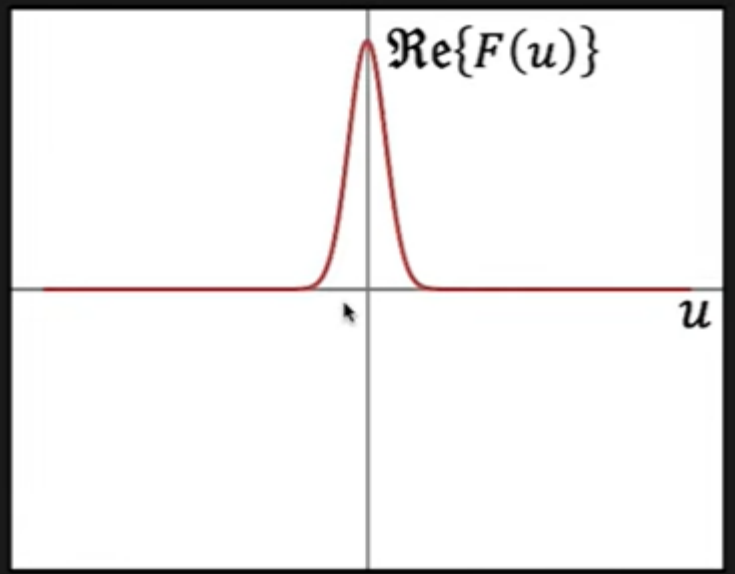
# Fourier Transform Examples

## Signal $f(x)$




$$f(x) = e^{-ax^2}$$

## Fourier Transform $F(u)$

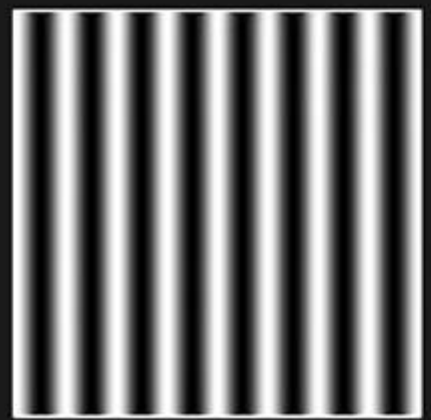



$$F(u) = \sqrt{\pi/a}\ e^{-\pi^2 u^2/a}$$

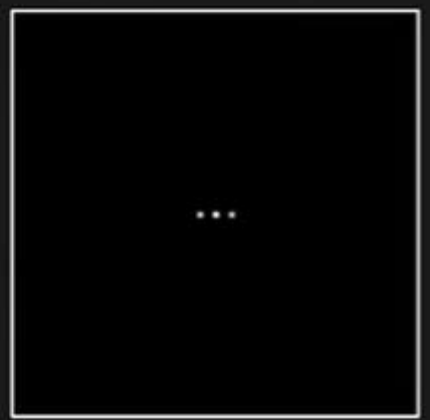
# Properties of Fourier Transform

| Property | Spatial Domain | Frequency Domain |
|---|---|---|
| Linearity | $\alpha f_1(x) + \beta f_2(x)$ | $\alpha F_1(u) + \beta F_2(u)$ |
| Scaling | $f(ax)$ | $\frac{1}{\lvert a \rvert} F\left(\frac{u}{a}\right)$ |
| Shifting | $f(x-a)$ | $e^{-i2\pi u a} F(u)$ |
| Differentiation | $\frac{d^n}{dx^n}(f(x))$ | $(i2\pi u)^n F(u)$ |

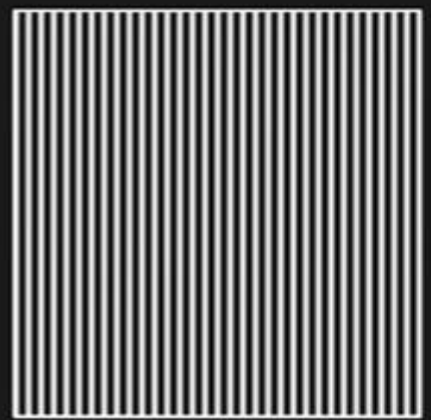
# 2D Fourier Transform: Example 1



$f(m,n)$



$\log(|F(p,q)|)$

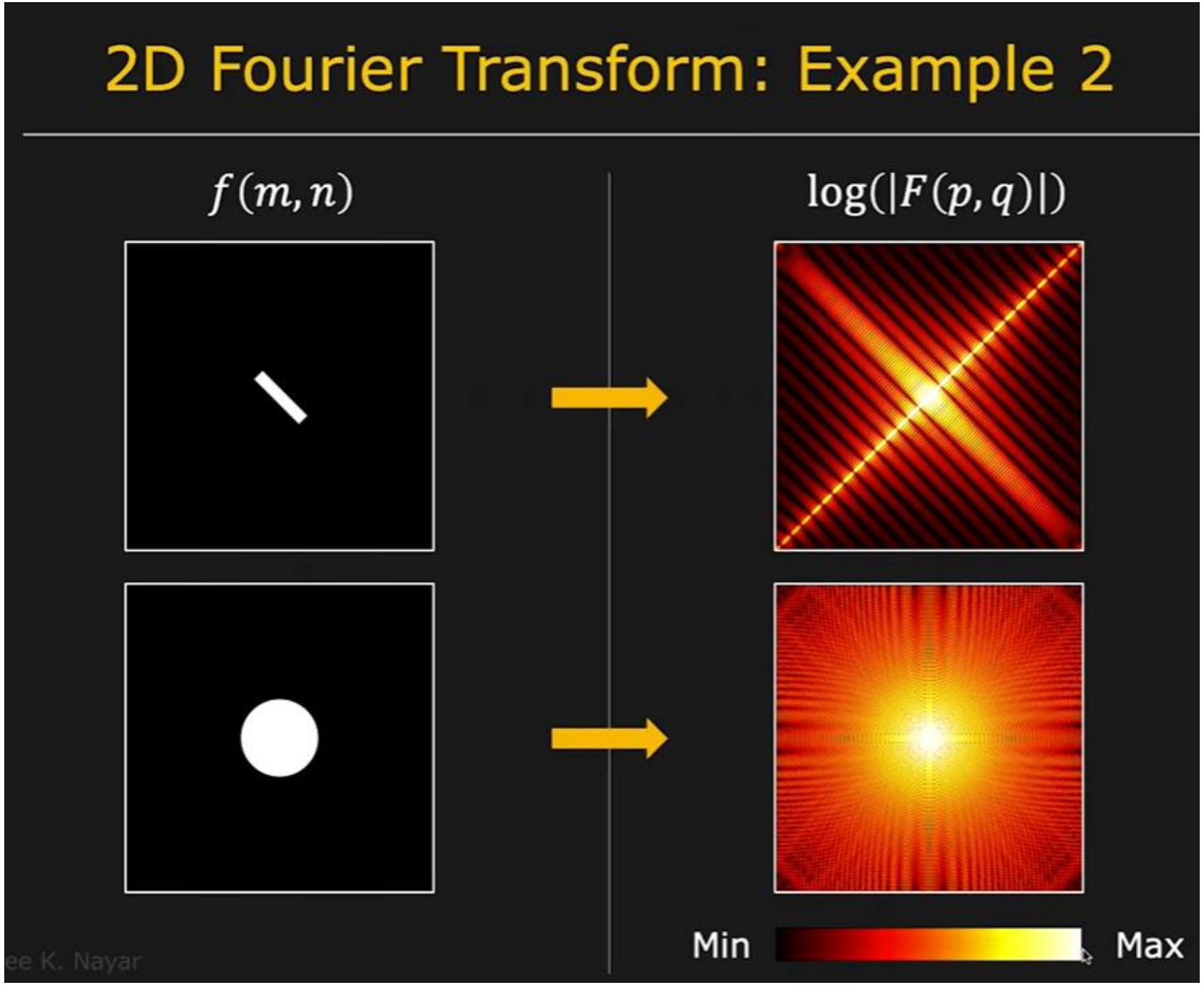


$g(m,n)$



$\log(|G(p,q)|)$

# 2D Fourier Transform: Example 2



# 2D Fourier Transform: Example 3

$f(m,n)$

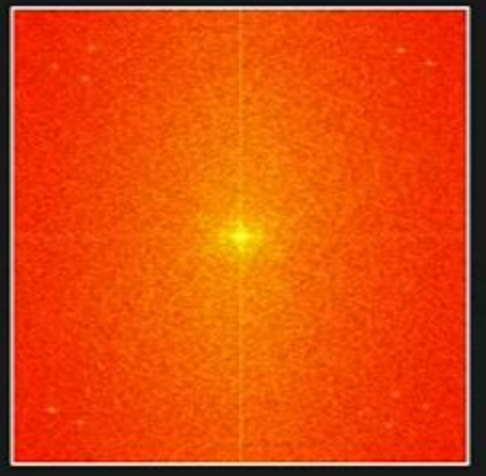
$\log(|F(p,q)|)$











Min



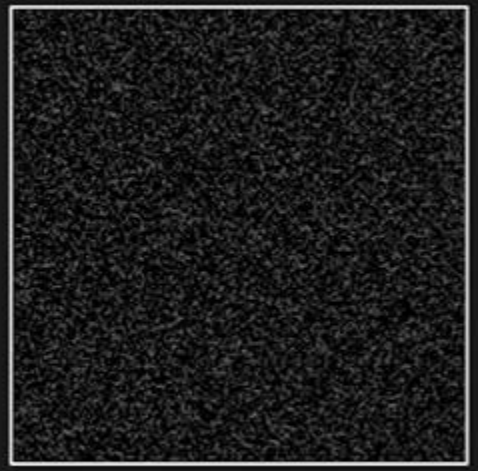
Max

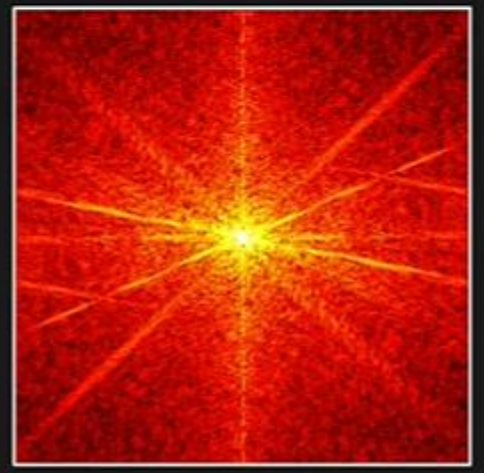
# 2D Fourier Transform: Example 4

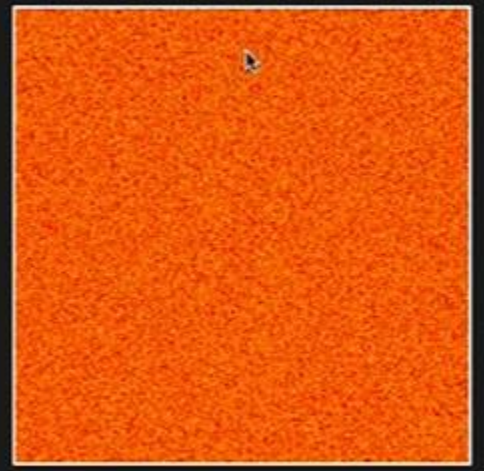
$f(m,n)$

$\log(|F(p,q)|)$









Min

Max

ee K. Nayar

# Low Pass Filtering

$f(m,n)$



$\log(|F(p,q)|)$



# Low Pass Filtering

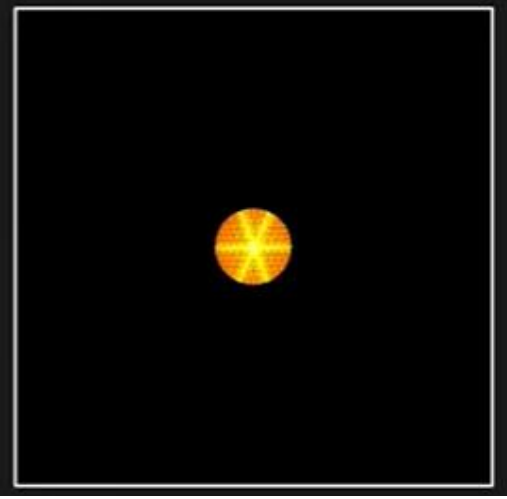
$f(m,n)$

$\log(|F(p,q)|)$









e K. Nay

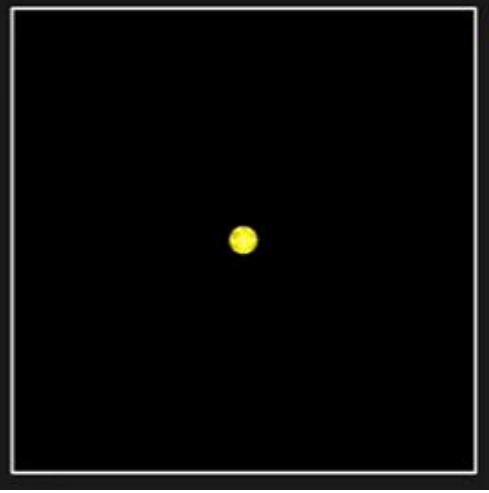
# Low Pass Filtering

$f(m,n)$



$\log(|F(p,q)|)$







ee K. Nav

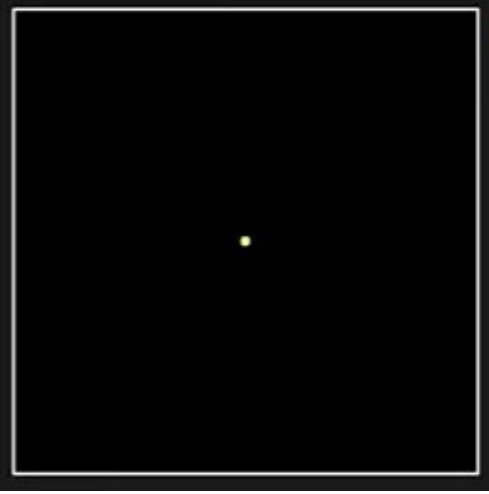
# Low Pass Filtering

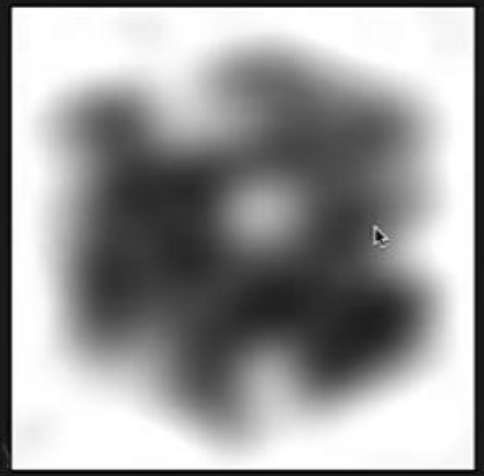
$f(m,n)$



$\log(|F(p,q)|)$







ee K. Nay

# High Pass Filtering

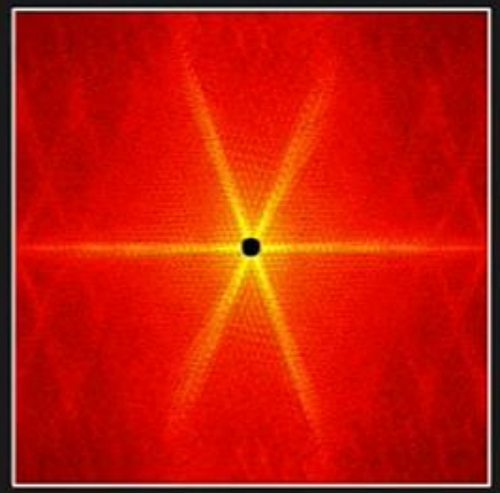
$f(m,n)$



$\log(|F(p,q)|)$







ee K. Nayar

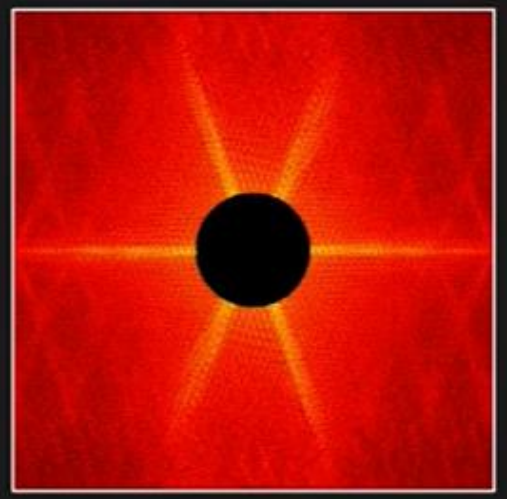
# High Pass Filtering

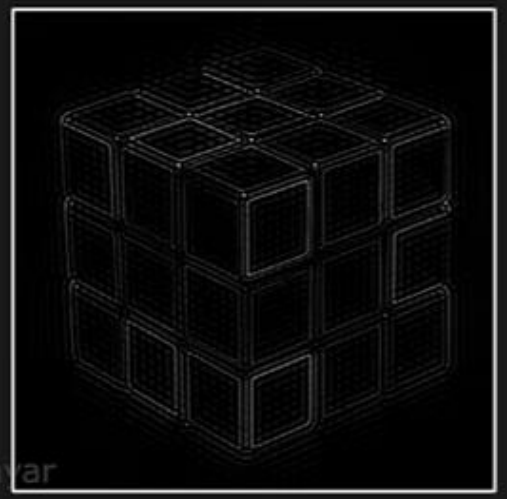
$f(m,n)$



$\log(|F(p,q)|)$

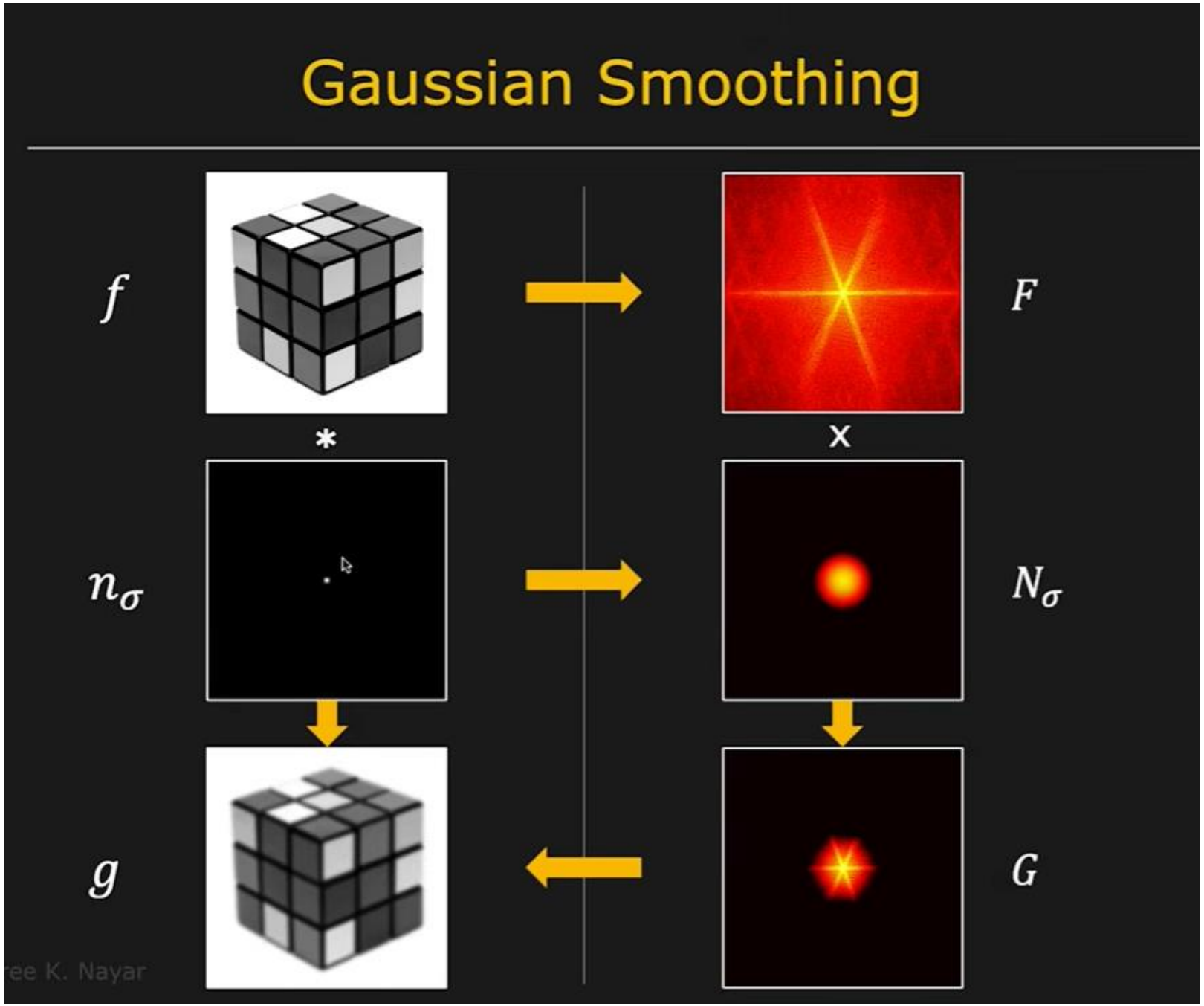






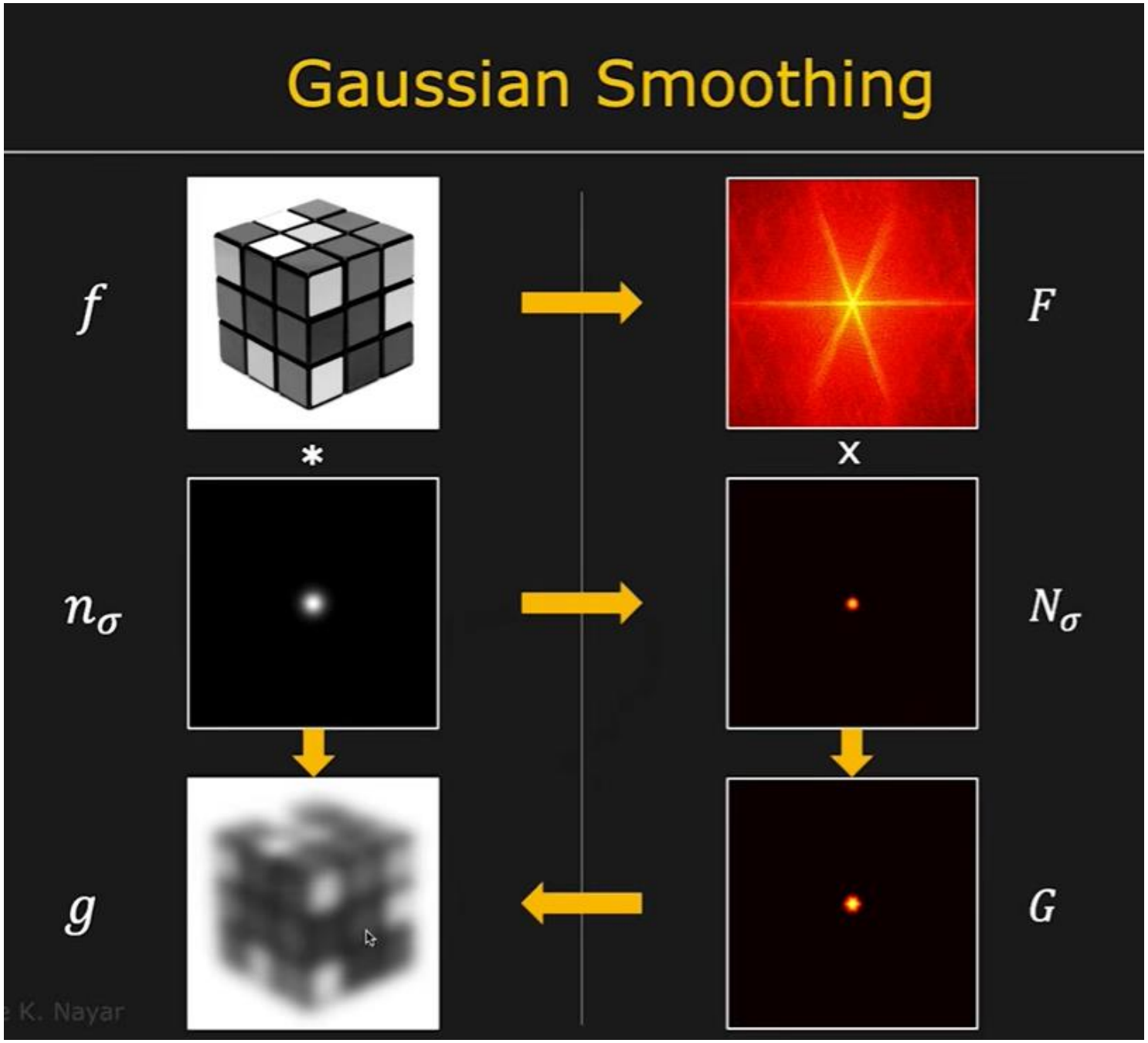
e K. Navar

# Gaussian Smoothing

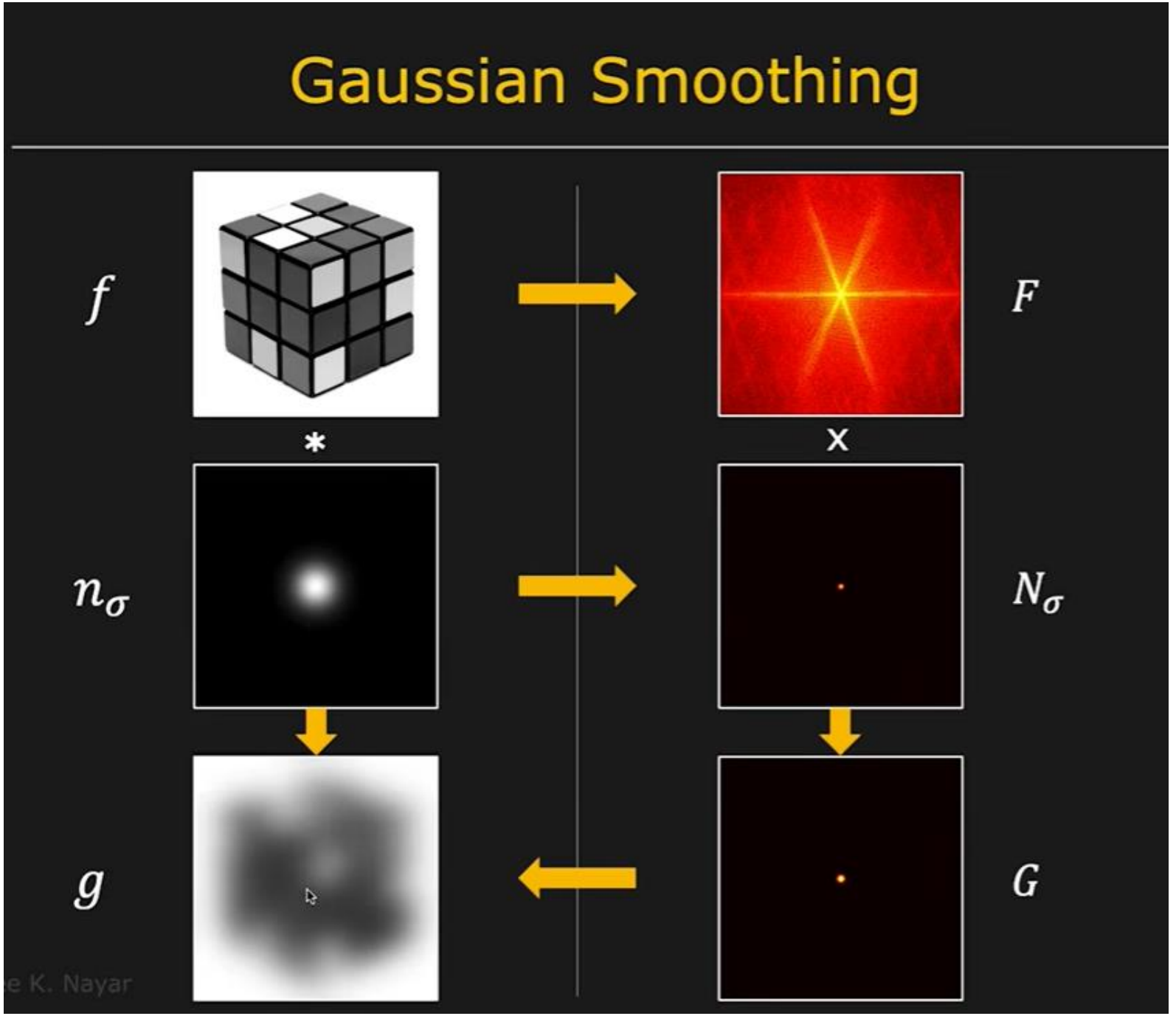



ree K. Nayar

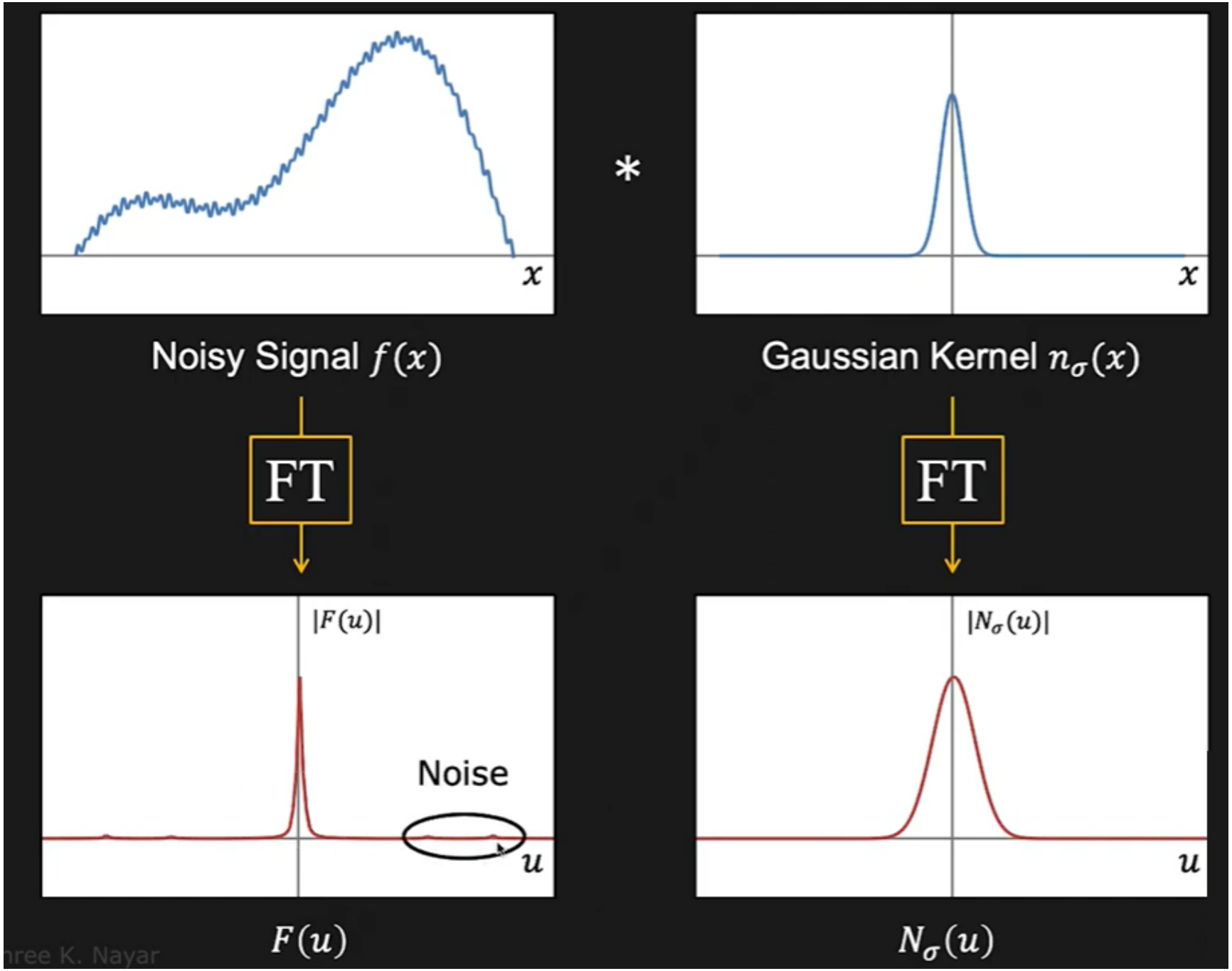
# Gaussian Smoothing

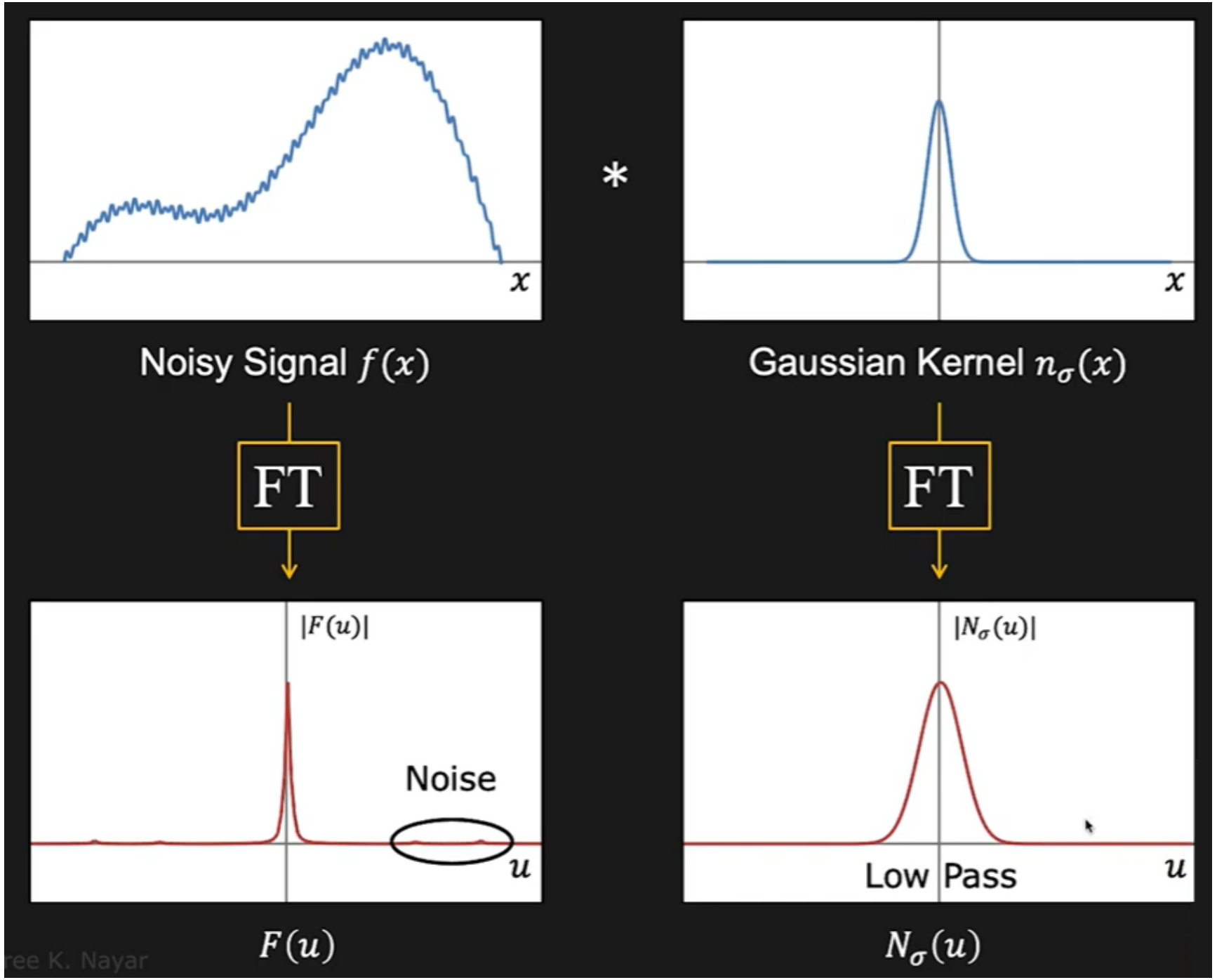


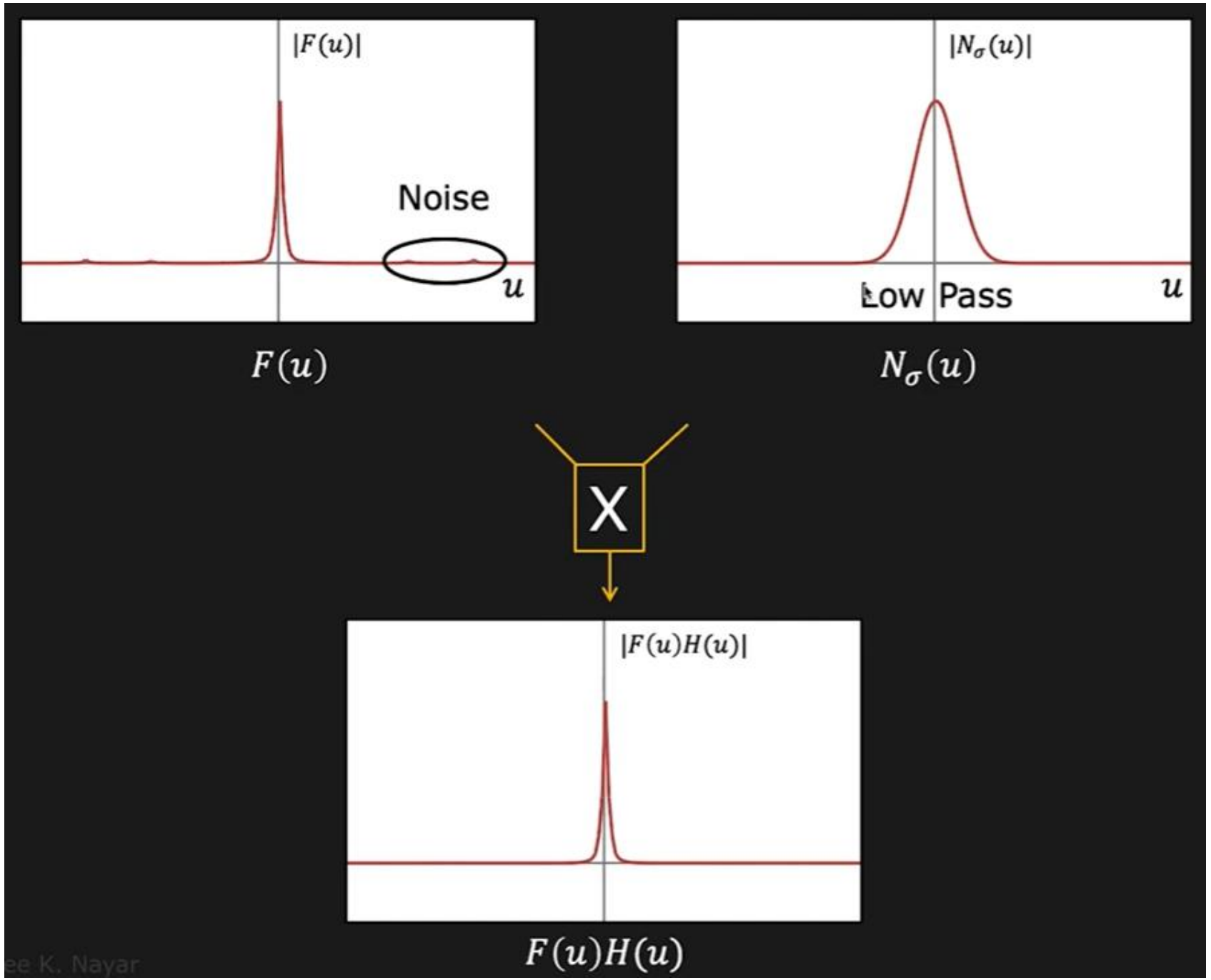
# Gaussian Smoothing

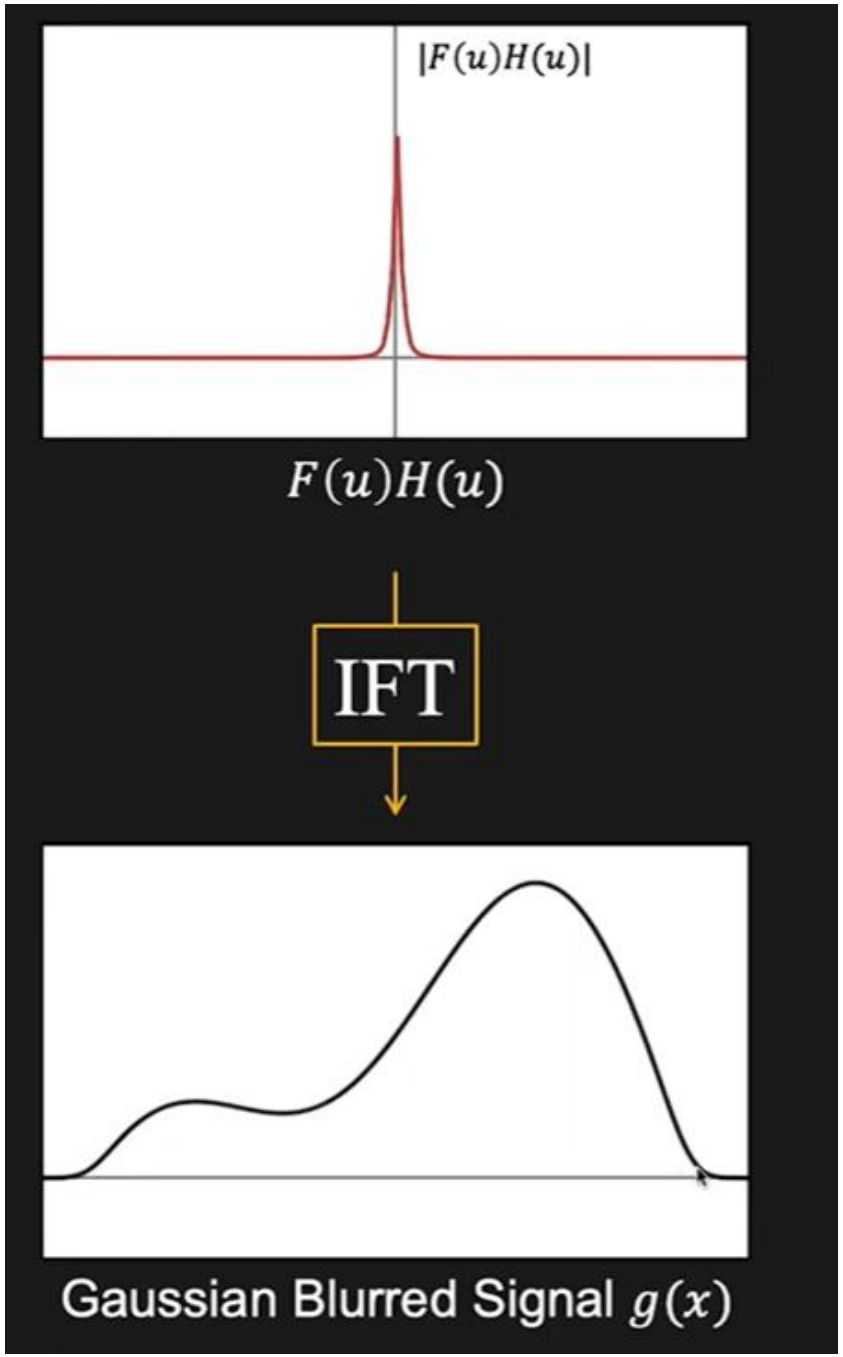


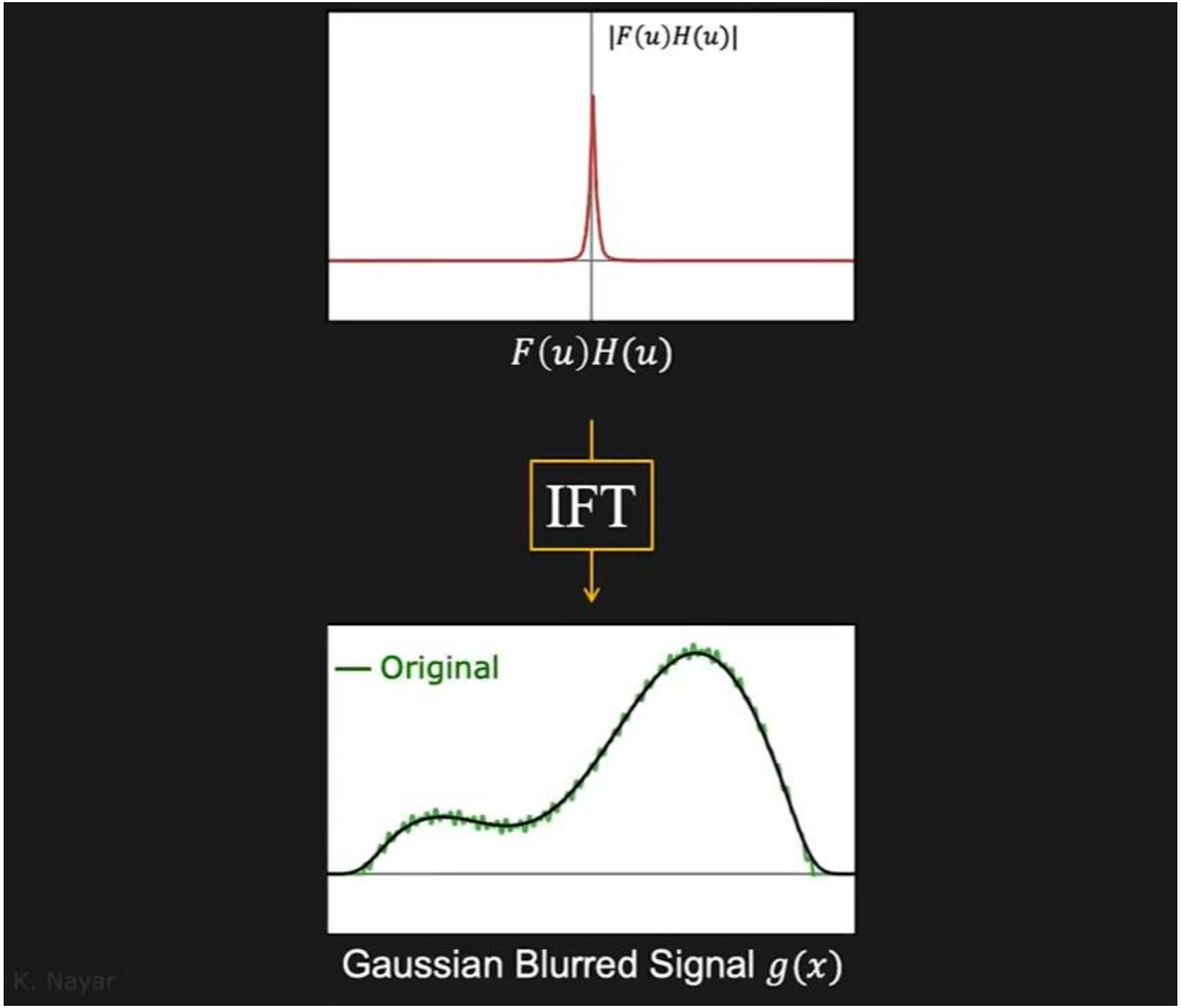

e K. Nayar


x
*
x
Noisy Signal $f(x)$
Gaussian Kernel $n_\sigma(x)$
FT
FT
$|F(u)|$
Noise
u
$|N_\sigma(u)|$
u
$F(u)$
$N_\sigma(u)$


x
*
x
Noisy Signal $f(x)$
Gaussian Kernel $n_\sigma(x)$
FT
FT
$|F(u)|$
Noise
u
$|N_\sigma(u)|$
Low Pass
u
$F(u)$
$N_\sigma(u)$


$|F(u)|$
Noise
$u$
$F(u)$
$|N_\sigma(u)|$
Low Pass
$u$
$N_\sigma(u)$
X
$|F(u)H(u)|$
$F(u)H(u)$

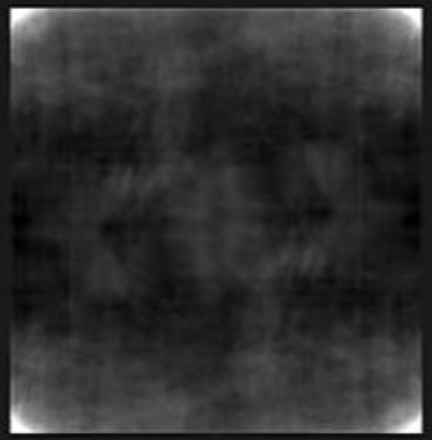

$|F(u)H(u)|$
$F(u)H(u)$
IFT
Gaussian Blurred Signal $g(x)$


$|F(u)H(u)|$
$F(u)H(u)$
IFT
— Original
Gaussian Blurred Signal $g(x)$
K. Nayar

# Importance of Phase



Original Image

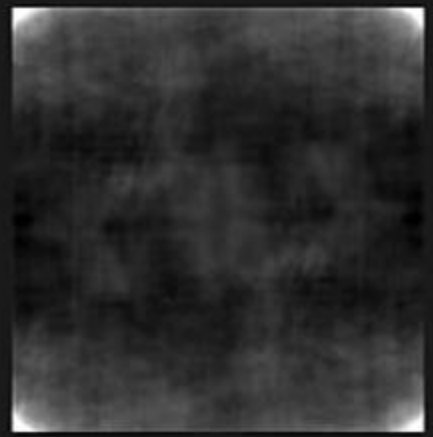


Magnitude Preserved,
Phase Set to Zero

# Importance of Phase



Original Image



Magnitude Preserved,
Phase Set to Zero

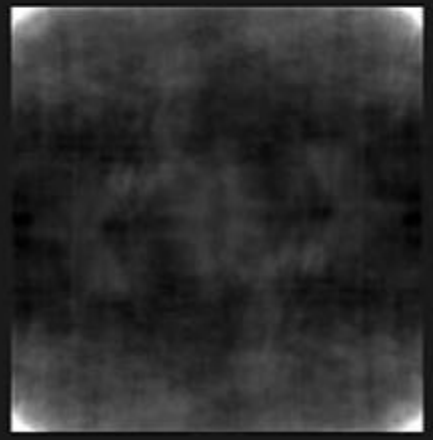


Phase Preserved,
Magnitude Set to Average
of Natural Images

# Importance of Phase



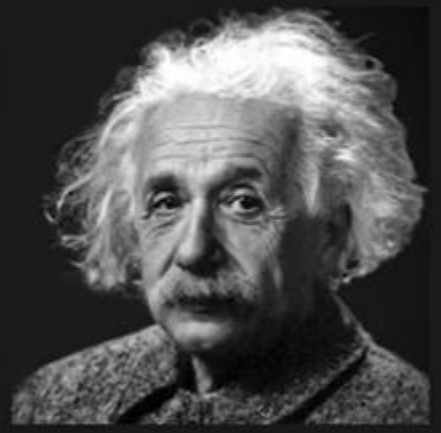
Original Image

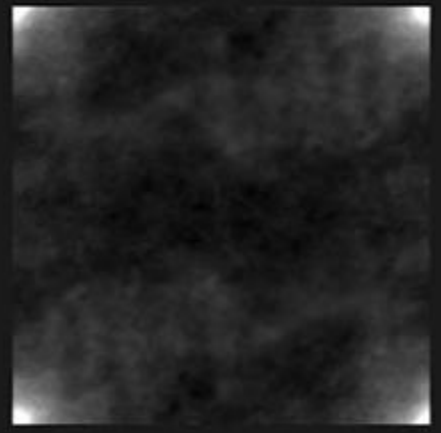


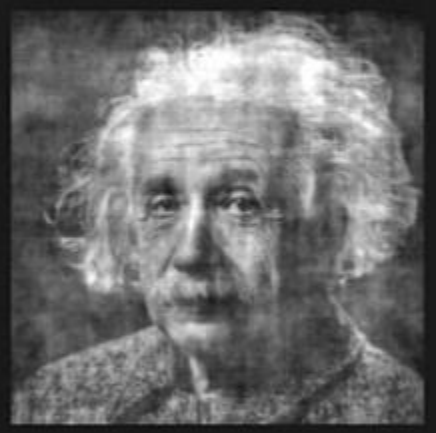
Magnitude Preserved,
Phase Set to Zero



Phase Preserved,
Magnitude Set to Average
of Natural Images





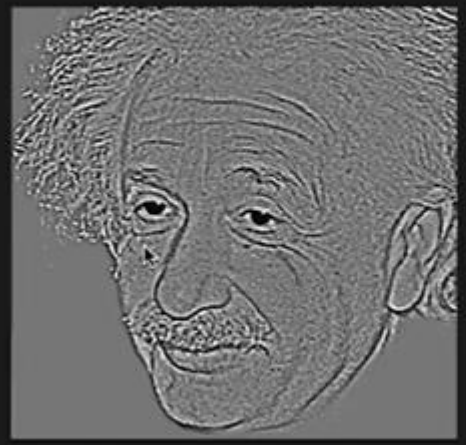


[Oppenheim

# Hybrid Images



Low Freq Only

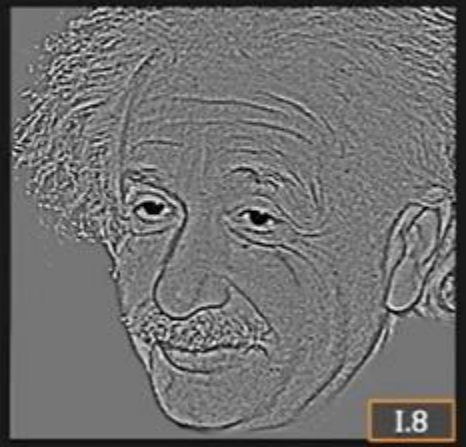


High Freq Only

# Hybrid Images



Low Freq Only




High Freq Only



Hybrid (Sum) Image

# References


- https://www.youtube.com/watch?v=kQUfKf-vPdI
- https://www.youtube.com/watch?v=tEzgtbnbXgQ
- https://www.youtube.com/watch?v=OOu5KP3Gvx0